

SIMATS ENGINEERING

ASSESSMENT TOOL 2

Course Code: ITA0606

Course Name: Machine Learning

Course Outcome Covered

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naive Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model (BL 6)

Assessment Tool Weightage: 20%

Questions: 15 Total Marks: 25

Annexure A – Case Study

Code of Conduct:

I **THYANESHWARAN** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

THYANESHWARAN

Annexure A – Short Answer Test

1. A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naive Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.
2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naive Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.
3. A retail company wants to segment its customers based on purchasing behavior, but the data contains hidden patterns and incomplete information. Apply the EM algorithm to identify clusters and estimate underlying distributions. Analyze convergence behavior, initialization challenges, and the role of probability learning. Discuss how the results can be used for targeted marketing and evaluate limitations of the approach.
4. A research team is developing a concept learning system to classify medical images into normal and abnormal categories. Using hypothesis space search and the Candidate Elimination approach, design a model that learns from labelled examples. Analyze the impact of finite vs infinite hypothesis spaces, sample complexity, and mistake bound model on learning performance. Evaluate how well the system generalizes to unseen data.
5. A financial institution needs to make investment decisions under uncertain market conditions. Using Bayesian inference, Gibbs algorithm, and probability learning, design a system to predict outcomes and update beliefs dynamically. Analyze how prior knowledge influences predictions, evaluate risk using posterior probabilities, and discuss how the model adapts as new data becomes available. Consider computational challenges and optimization strategies.

Rubrics for Calculation

Criteria	Wt	Level 4 (5)	Level 3 (4)	Level 2 (2-3)	Level 1 (0-1)
Conceptual Understanding	30	Clear & complete understanding	Good, minor gaps	Basic, some misconceptions	Poor or incorrect
Accuracy of Answer	20	Completely correct	Mostly correct, minor errors	Partially correct, noticeable errors	Incorrect / irrelevant
Application of Knowledge	20	Correctly applies concepts	Minor mistakes	Limited / partially correct	No / incorrect application
Completeness of Response	15	Covers all required parts	Minor omissions	Covers only some parts	Incomplete / missing
Clarity & Presentation	15	Clear, concise, well-organized	Minor issues	Somewhat unclear	Difficult to understand

Q1. Bayesian Framework for Intelligent Disease Diagnosis

Problem Setup: A hospital wants a diagnostic support tool that reasons under uncertainty, using patient symptoms and medical history to estimate the likelihood of various diseases.

Model Design: Bayes' Theorem forms the mathematical core: for each candidate disease D and observed symptom set S , the posterior probability $P(D|S)$ is computed from $P(S|D)$, the prior $P(D)$, and the evidence $P(S)$. A Naive Bayes classifier operationalizes this efficiently by assuming symptoms are conditionally independent given the disease, which keeps computation tractable even with many symptoms. Where symptoms are known to influence each other -- for example fever raising the likelihood of reported chest pain -- a Bayesian Belief Network is layered on top, encoding a directed graph of conditional dependencies so the model captures interactions that Naive Bayes would otherwise miss.

Handling Uncertainty: Because no single symptom is deterministic, every inference is expressed as a probability distribution rather than a hard decision. As new symptoms or test results arrive, the posterior is recalculated, letting belief in a diagnosis strengthen or weaken incrementally rather than committing early to a single answer.

Role of Conditional Dependencies: The BBN's structure means that evidence for one symptom propagates through the network to update belief in correlated symptoms and diseases, giving more realistic diagnoses than an independence assumption alone, particularly for diseases that share overlapping symptom profiles.

Probability Learning: Rather than being fixed by hand, the conditional probability tables and priors are learned from historical patient records, so the system's estimates improve and adapt as more diagnosed cases are added to the training data.

Real-World Constraints and Evaluation: In practice, missing fields in patient charts, inconsistent record-keeping, and shifting disease prevalence across seasons or populations all degrade accuracy. Regularly re-estimating priors from recent hospital data and combining Naive Bayes' speed with BBN's structural accuracy gives a system that is both fast enough for clinical use and explainable enough for a doctor to trust its reasoning.

Q2. Spam Detection System using Bayesian Classification

Problem Setup: The goal is to automatically separate incoming email into spam and legitimate (ham) categories using statistical features of each message.

System Design: Each email is converted into a feature vector -- presence of specific keywords, sender reputation, number of embedded links, and formatting cues. A Naive Bayes classifier estimates $P(\text{Spam}|\text{features})$ using the independence assumption across features, which works well in practice despite features rarely being truly independent. The Bayes Optimal Classifier is used as a theoretical benchmark: it combines predictions from every hypothesis in the space weighted by its posterior probability, giving the lowest possible error rate, though it is generally too expensive to compute directly and so Naive Bayes serves as a practical approximation.

MLE vs MDL: Maximum Likelihood Estimation picks the parameters that make the observed training emails most probable, which can lead to overfitting if the model chases noise in the training set. The Minimum Description Length principle instead balances how well the model fits the data against how simple the model is, penalizing unnecessary features and favoring a more compact classifier that generalizes better to unseen emails.

Performance and Trade-offs: Larger, more diverse training samples reduce variance in the estimated probabilities and improve accuracy. However, an overly large hypothesis space -- for instance considering every possible word combination -- increases both computational cost and the risk of overfitting; MDL-guided feature selection keeps the space manageable.

Conclusion: Naive Bayes gives an efficient, easily-updated spam filter, while MDL-based feature selection keeps the model simple enough to generalize well as spam tactics evolve over time.

Q3. Customer Segmentation using the EM Algorithm

Problem Setup: The retailer's purchasing data contains latent customer segments that are not directly labeled, along with incomplete records, making this a hidden-variable estimation problem.

Method: The Expectation-Maximization algorithm is applied by first assuming a mixture model (for example a mixture of Gaussians over spending features). The E-step computes the expected cluster membership probability for each customer given the current parameter estimates, and the M-step re-estimates the distribution parameters (means, variances, mixing weights) to maximize the expected likelihood. These two steps alternate until the parameters converge.

Convergence and Initialization: EM is guaranteed to improve (or maintain) the likelihood at every iteration, but it only finds a local optimum, so poor initial parameter guesses can trap it in a suboptimal clustering. Running EM multiple times from different random initializations and keeping the best result mitigates this risk.

Role of Probability Learning: Because EM works directly with probability distributions, it naturally handles the missing or incomplete purchase records by treating them as latent variables to be estimated rather than requiring complete data up front.

Application and Limitations: The resulting clusters -- for example high-value, occasional, and discount-driven shoppers -- can drive targeted marketing campaigns and personalized offers. The main limitations are sensitivity to initialization, the need to pre-specify the number of clusters, and the assumption of a particular distributional form, which may not match real purchasing behavior perfectly.

Q4. Concept Learning for Medical Image Classification

Problem Setup: Labelled medical images (Normal / Abnormal) are used to train a concept-learning system that searches a hypothesis space of possible classification rules over image features.

Model Design: The Candidate Elimination algorithm maintains two boundary sets: the Specific boundary (S), the most specific hypotheses consistent with observed positive examples, and the General boundary (G), the most general hypotheses consistent with observed negative examples. As each new labelled image arrives, S is generalized just enough to cover new positives and G is specialized to exclude new negatives, so the version space -- all hypotheses consistent with the data seen so far -- shrinks over time.

Finite vs Infinite Hypothesis Spaces: When the hypothesis space is finite, sample complexity bounds guarantee convergence to a good hypothesis within a bounded number of examples. Infinite hypothesis spaces (such as continuous-valued image features) require tools like the VC dimension to bound the number of examples needed, and generally demand more data before the version space narrows sufficiently.

Mistake Bound Model: The mistake bound framework measures how many prediction errors a learner makes before converging on the correct concept; providing more informative, diverse training examples early reduces the number of mistakes made during learning.

Generalization: A balanced dataset across both classes, combined with representative and discriminative image features, allows the learned version space to generalize reliably to new, unseen images rather than overfitting to the training set.

Conclusion: Candidate Elimination offers an interpretable, incrementally-updatable concept learner, and its performance is directly governed by how quickly the hypothesis space can be narrowed given the available labelled data.

Q5. Bayesian Inference for Investment Decision-Making

Problem Setup: A financial institution must estimate future market outcomes despite incomplete and noisy information, making this a natural fit for Bayesian probabilistic modeling.

System Design: Bayesian inference is used to maintain a probability distribution over possible market states, updating it as new data (price movements, economic indicators) arrives via Bayes' Theorem. When the posterior distribution cannot be computed in closed form -- common in high-dimensional financial models -- the Gibbs sampling algorithm approximates it by iteratively sampling each variable conditioned on the current values of the others, eventually producing samples that reflect the true posterior.

Role of Priors and Posteriors: Early in the process, predictions rely heavily on prior beliefs (e.g. historical market behavior or expert judgment). As more market data is observed, the posterior increasingly dominates, so the model's predictions become progressively more data-driven and less dependent on the initial assumptions.

Risk Evaluation: Because the output is a full posterior distribution rather than a single point estimate, the institution can quantify uncertainty directly -- for instance by examining the spread of the posterior to assess the probability of adverse outcomes, supporting more informed risk management than a purely deterministic forecast.

Computational Challenges: High-dimensional market data makes exact inference intractable, so Gibbs sampling's iterative approach trades exact computation for tractable approximation; techniques such as parallelized sampling and efficient convergence diagnostics help keep runtime practical.

Conclusion: A Bayesian inference pipeline, supported by Gibbs sampling, lets the institution continuously refine its market beliefs and make risk-aware investment decisions as new information becomes available.